

Rewriting Numbers in Equivalent FormsWorksheet 7A.01.01 **ANSWER KEY**

1. Which strategy will always work to rewrite a fraction as a decimal? Fill in one bubble.

- ☐ Multiply or divide by 100.
- ☒ **Divide the numerator by the denominator.**
- ☐ Create an equivalent fraction with a denominator of 100.
- ☐ Use the place value of the last digit to determine the denominator.

Dividing always works. Making an equivalent fraction with a denominator of 100 works for $17/20$, but not for $1/3$ — no whole number times 3 equals 100. Multiplying or dividing by 100 moves between decimals and percentages, not fractions.

2. Which strategy will always work to rewrite a terminating decimal as a fraction? Fill in one bubble.

- ☐ Multiply or divide by 100.
- ☐ Divide the numerator by the denominator.
- ☐ Create an equivalent fraction with a denominator of 100.
- ☒ **Use the place value of the last digit to determine the denominator.**

In 0.85 the last digit is in the hundredths place, so the denominator is 100 and the fraction is $85/100$, which simplifies to $17/20$. A denominator of 100 works for any decimal that stops at or before the hundredths place — 0.8 is $80/100$ as well as $8/10$.

For questions 3 through 5, write each fraction as an equivalent decimal. Use bar notation for any decimal that repeats. Then fill in one bubble.

3. $\frac{5}{6} = 0.8\overline{3}$

This is a ☐ terminating decimal ☒ **repeating decimal.**

4. $\frac{17}{8} = 2.125$

This is a ☒ **terminating decimal** ☐ repeating decimal.

5. $\frac{14}{3} = 4.\overline{6}$

This is a ☐ terminating decimal ☒ **repeating decimal.**

$5 \div 6 = 0.8333\dots$, so only the 3 repeats. $17 \div 8 = 2.125$ exactly, so it terminates. $14 \div 3 = 4.666\dots$, so the 6 repeats.

6. The decimal equivalent of $28/15$ is $1.8666\dots$. Which of these is $1.8666\dots$ written in bar notation? Fill in one bubble.

- ☐ A. $1.\overline{86}$
- ☐ B. $1.\overline{8}6$

- C. $1.8\overline{6}$
- D. 1.86

Only the 6 repeats: $28 \div 15 = 1.8666\dots$, so the bar goes over the 6 alone. B would mean $1.868686\dots$ D is 1.86 exactly, a different number. A puts a bar over the whole number, which bar notation never does.

7. $\frac{1}{9} = 0.\overline{1}$
 $1 \div 9 = 0.111\dots$

Use your answer to question 7 for questions 8 through 10. Predict each decimal equivalent, then check it with your calculator. Use bar notation for every repeating decimal.

8. $5\frac{4}{9} = 5.\overline{4}$

9. $\frac{7}{9} = 0.\overline{7}$

In ninths the numerator is the digit that repeats. $4/9 = 0.444\dots$, so $5\frac{4}{9} = 5.444\dots$ $7/9 = 0.777\dots$

10. Use $101/9$ for parts a and b.

a. Write $101/9$ as a mixed number. $11\frac{2}{9}$

b. Write $101/9$ as a decimal. Use bar notation. $11.\overline{2}$

$101 \div 9 = 11$ with a remainder of 2, so $101/9 = 11\frac{2}{9}$. The ninths pattern gives $2/9 = 0.222\dots$, so the answer is $11.222\dots$

11. Priya divides 5 by 9 on her calculator. The screen shows 0.555555556 .

a. Is $5/9$ a terminating decimal or a repeating decimal? Fill in one bubble.

- Terminating.
- **Repeating.**

b. Write $5/9$ in bar notation. $0.\overline{5}$

c. Explain why the last digit on the screen is a 6.

The calculator has room for only ten digits, so it rounds the last one it shows. The true value is $0.5555\dots$ and it never stops, so the 6 on the screen is the calculator rounding, not part of the pattern.

Read the pattern from the middle of the display, never from the last digit.